# Coordination-Collapse Phase Boundary in Policy-Gradient Cooperative MARL

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## Abstract

Cooperative multi-agent reinforcement learning (MARL) can collapse into a degenerate *lazy-agent* policy when reward centralization is too low, but this failure has only been described qualitatively. We make it measurable: with a pre-registered collapse definition and bootstrap confidence intervals, we estimate the critical centralization $w_{\text{crit}}$ at which coordination gives way to collapse. Using this boundary as an instrument, we find that behavioral *loss aversion* ($\lambda > 1$) shifts it rightward—loss-averse agents demand more centralization to coordinate. The direction is monotone in $\lambda$, its sign is fixed a priori by a mean-field mechanism, and it replicates across four environment–algorithm settings; the boundary's metastable slide with the training horizon likewise generalizes. The instrument then exposes two findings: prospect theory's two components move the boundary in *opposite* directions, and—on a directed-flow substrate—a single *downstream* loss-averse agent can set the whole system's requirement, where the average agent would not. We report what generalizes and what is single-substrate, and frame the directed-flow cascade as a falsifiable prediction.


## Introduction

A team of learning agents sharing a task can fall into a trap their designers never intended: instead of coordinating, every agent quietly defaults to inaction, and the team settles into a stable, useless equilibrium. Cooperative MARL calls this *coordination collapse*, with the *lazy-agent* policy as its endpoint (Liu et al. 2023; Shi and Peng 2023)—an instance of the sequential social dilemma that arises whenever private incentives drift from the collective good (Leibo et al. 2017). Whether a team falls in is governed by one design dial, the *reward centralization* $w \in [0, 1]$: the share of each agent's reward tied to the team's outcome rather than its own. Turn $w$ down and an agent's private performance drowns out its contribution, so cooperating earns no learnable signal and the safe policy wins. The field knows this only as a direction—*more centralization helps* (Akella 2025; Liu et al. 2023)—and stops there. *Where*, in $w$, a working team tips into collapse has never been measured. The nearest work mitigates the lazy-agent pathology without locating a boundary (Liu et al. 2023; Shi and Peng 2023; Weng and Lee 2026), or sweeps neighbouring axes—credit-assignment level (Zhao and Xie 2025), reward weights (Jiang et al. 2020). Closest in spirit, Yamaguchi (2025) build a phase diagram for independent Q-learning over *environmental* axes (grid size, density), and Capannoli et al. (2026) prove a metastable boundary in network opinion dynamics under a fixed update rule; neither measures a reward-structure threshold in gradient-trained cooperative policies, with confidence intervals, that a behavioral bias then moves.

Pinning that number down matters now, because the agents being deployed are no longer neutral optimizers. In automated replenishment, fraud review, and energy dispatch, learning agents act for or beside people and absorb human preference structure by construction—whether trained on human choices through preference-aligned objectives that are themselves prospect-theoretic in form (Ethayarajh et al. 2024), or sharing the loop with human operators (Lalmohammed et al. 2025). The sturdiest such regularity is *loss aversion*: a loss looms larger than an equal gain (Kahneman and Tversky 1979; Tversky and Kahneman 1992). Behavioral economics knows what this does to one chooser; it is silent on what it does to a *group* learning to coordinate (Leibo et al. 2017; Lowe et al. 2017). A measured collapse boundary is the instrument that question needs, and it extends a thin existing bridge between behavioral preferences and collective learning dynamics—inequity aversion in social dilemmas (Hughes et al. 2018), deterministic limits of multi-agent learning (Barfuss, Donges, and Kurths 2019; Hu, Leung, and Leung 2019)—to the prospect-theoretic preferences that today's training pipelines instill.

We therefore build the ruler first, and treat it as the paper's central object. Following the posture physics takes toward a phase boundary, we fix a binary collapse criterion in advance, freeze it, fit $P(\text{collapse} \mid w)$,

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and bootstrap confidence intervals on $w_{\text{crit}}$. (We mean "phase boundary" empirically—a sharp, metastable threshold whose $T \to \infty$ limit is the boundary proper—not a thermodynamic singularity.) Two design choices keep the readings honest: we apply cumulative prospect theory (CPT) as a *reward transform* after blending rather than inside the update, so the optimizer is bit-identical across conditions and any movement of $w_{\text{crit}}$ is behavioral; and we re-measure at several horizons, since a drifting threshold could pass for a constant. Our contributions are: (i) the first confidence-intervalled phase boundary of coordination collapse in deep cooperative MARL, with a reusable pre-registered protocol; (ii) the finding that loss aversion shifts it rightward, a direction whose sign we fix a priori and confirm across four environment–algorithm settings, with the metastable horizon slide generalizing likewise; (iii) a decomposition showing prospect theory's two components move it in opposite directions; and (iv) the result that a downstream loss-averse *minority* can set the boundary on a directed-flow substrate. Our novelty is narrow: Mengesha, Sun, and Roy (2025) show loss aversion moves a tipping point, but in evolutionary game theory over fixed strategies, and Lalmohammed et al. (2025) bring CPT into cooperative MARL without separating its components or measuring a boundary. Ours is the measured, CI'd boundary and the component decomposition against it, in gradient-trained neural policies.

## Preliminaries and Method

In cooperative MARL, $N$ agents share an environment and maximize a joint objective under CTDE (Lowe et al. 2017); the reward-mixing weight $w \in [0, 1]$ interpolates purely local ($w$=0) and purely global ($w$=1) reward. As $w \to 0$ the cooperative signal is diluted and agents collapse to the safe lazy policy (Liu et al. 2023; Shi and Peng 2023). This $w$ is our sole structural intervention; all else is held fixed, so any boundary shift is attributable to $w$ and the behavioral parameters. Cumulative prospect theory (Kahneman and Tversky 1979; Tversky and Kahneman 1992) gives the value function $v(x) = x^{\alpha}$ for $x \geq 0$ and $-\lambda(-x)^{\beta}$ for $x < 0$, where $\alpha, \beta \in (0, 1]$ govern curvature and $\lambda > 1$ is loss aversion ($\alpha$=$\beta$=1, $\lambda$=1 is linear utility). We apply it as a reward transform after blending,

$$r_i^{\text{PT}} = v\big((1-w)\, r_i^{\text{local}} + w\, r^{\text{global}}\big)\,, \qquad (1)$$

the decisive difference from Lalmohammed et al. (2025), who fold CPT *into* the actor–critic update, entangling curvature with loss aversion and confounding any shift with a change to the optimizer. Transforming only the scalar reward leaves the policy gradient identical across conditions, so a measured shift is behavioral and the two channels dial independently. We pre-registered (commit `03c3fc6`) the binary collapse label $\mathbf{1}[\bar{S} < 10 \wedge \bar{\pi} \leq -127.1]$ ($\bar{S}$ mean base-stock, $\bar{\pi}$ mean profit), robust to ±20% threshold perturbation (max $\Delta w_{\text{crit}} = 0.015$; Supplement). For each condition we fit $P(\text{collapse} \mid w) = \sigma(b_0 + b_1 w)$, estimate $w_{\text{crit}} = -b_0/b_1$, and bootstrap its 95% CI; steep high-$\lambda$ cells use Firth penalized regression, which agrees with the bootstrap throughout. The decomposition compares three conditions with identical environment and optimizer: **Null** ($\alpha$=$\beta$=1, $\lambda$=1), **Curvature** ($\alpha$=$\beta$=0.88, $\lambda$=1), and **Loss aversion** ($\alpha$=$\beta$=0.88, $\lambda > 1$). The primary substrate is `NetworkManagement-v1` (Hubbs et al. 2020), a six-node multi-echelon inventory network; agents run PPO (Schulman et al. 2017) (20 seeds/condition, 1,524 primary runs). Full setup and hyperparameters are in the Supplement.

## The Boundary and Its Behavioral Shift

**A sharp, metastable boundary.** Collapse is a sharp logistic transition in $w$, not a gradual decay: at 300k steps $w_{\text{crit}} = 0.666$ $[0.635, 0.694]$ (transition width $\Delta w = 0.199$, $p = 3.2 \times 10^{-7}$), with collapse fractions rising from near-zero to near-one within $\Delta w < 0.3$ (Figure 1a). The boundary is not a constant: it slides left with the training horizon, $w_{\text{crit}} = 0.822$ at 150k to 0.666 at 300k. Intermediate-$w$ collapse is therefore a *metastable* attractor, not an absorbing state. This slide is not a supply-chain artifact: under centralized-critic MAPPO it recurs in both transfer environments—on MPE $w_{\text{crit}}$ falls $0.82 \to 0.68$ as the horizon grows 2M → 8M, and on Level-Based Foraging $0.33 \to 0.20$ over 1M → 4M, both with non-overlapping intervals (Figure 1b). The two supply-chain horizons anchor the $-\ln T$ trend the mechanism predicts; we did not estimate the asymptote, and additional matched horizons are in progress.[1] A basin-control experiment rules out initialization as the driver (Supplement).

**Loss aversion raises the boundary; curvature lowers it.** Against a single canonical null—$w_{\text{crit}}^{\text{null}} = 0.666$ $[0.635, 0.694]$, independently reproduced at 0.685 by a dedicated experiment (every other null-like number is a labeled offset, not a separate baseline)—we sweep $\lambda \in \{1, \ldots, 7\}$ ($\alpha$=$\beta$=0.88, 300k). The boundary rises monotonically with loss aversion—$\{0.586, 0.715, 0.774, 0.806, 0.839, 0.836, 0.851\}$—then saturates near 0.85 with clear headroom to the $w$=1 ceiling, so the plateau is behavioral, not a ceiling artifact (Figure 2, Table 1); the endpoints are non-overlapping by 0.221. Curvature acts in the *opposite* direction, *lowering* $w_{\text{crit}}$ to 0.586 ($-0.100$ $[-0.172, -0.033]$ vs. the null). That the two components of a single "risk-averse" psychology pull the boundary in opposite directions is the first of two surprises, and is visible only because there is a calibrated null to read each component against.

**Two ways to count the loss-aversion effect.** The magnitude depends on the baseline, and we report both explicitly to avoid ambiguity (Table 1). Measured

[1] Runs at 450k and 900k ($\alpha$=$\beta$=1) are queued to convert the asserted $T \to \infty$ trend into a fitted one; they are not in this version.

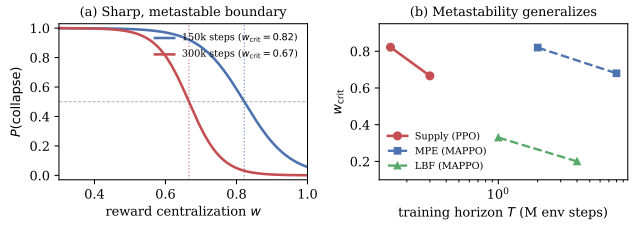



Figure 1: (a) Collapse is a sharp logistic transition in $w$; the boundary sharpens and slides left from 150k to 300k steps. (b) The metastable slide generalizes: $w_{\text{crit}}$ falls with the training horizon on the supply chain (PPO) and on both transfer environments under MAPPO (MPE, LBF), each with non-overlapping endpoint intervals.

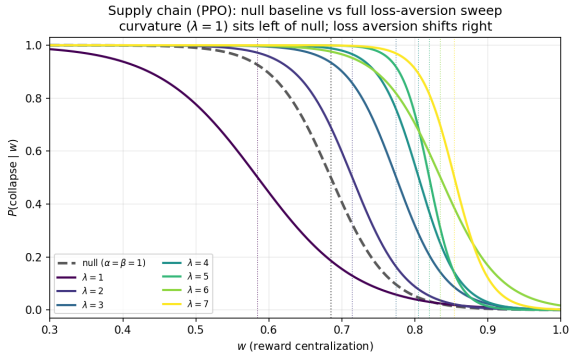



Figure 2: Behavioral decomposition and full loss-aversion sweep (supply chain, PPO). Curves are Firth-fitted $P(\text{collapse} \mid w)$; dotted lines mark $w_{\text{crit}}$. Curvature ($\lambda$=1) sits *left* of the null (easing coordination); loss aversion ($\lambda$=2, . . . , 7) shifts $w_{\text{crit}}$ *right*, rising then saturating.

*against the linear null* (0.685), the combined CPT shift is +0.089 [0.041, 0.141] at $\lambda$=3 and +0.166 [0.135, 0.202] at $\lambda$=7—but this nets loss aversion against curvature's −0.100. The *pure loss-aversion increment*, measured against the curvature baseline ($\lambda$=1, 0.586)—the correct isolate, since it holds curvature fixed—is larger: +0.188 at $\lambda$=3 and +0.265 at $\lambda$=7. The two are reconciled by the curvature offset ($-0.100 + 0.265 = 0.165 \approx 0.166$). Throughout, "the loss-aversion effect" refers to the increment over the curvature baseline; the net-vs-null figure is the deployment-relevant total. The full sweep is tracked by a one-parameter, theory-bounded affine law (§) to RMSE 0.014; high-$\lambda$ cells are steep and rest on a single interior grid point, so a finer $w \in \{0.80, 0.90\}$ grid (queued) would tighten those intervals.

## A Mean-Field Mechanism Fixes the Sign

We give a mechanistic account of why the boundary exists and which way each component moves it. Following deterministic-limit and mean-field treatments of multi-agent learning (Barfuss, Donges, and Kurths 2019; Hu, Leung, and Leung 2019; Hussain, Belardinelli, and Leonardos 2025), we model each Gaussian-policy PPO agent's mean action as a Langevin variable; the machinery is standard and is in the Supplement. *We claim the mechanism at the level of sign and functional form, not magnitude.* Modelling the policy mean as $\mathrm{d}\mu_i = D(w)\,\mathrm{d}t + \sqrt{2\Theta(w)}\,\mathrm{d}W$, with drift $D$ and a centralization-weighted diffusion $\Theta$ into which partner exploration enters, the replenishing regime makes an upstream node's inventory a local cost but a global benefit, giving cooperative drift $D(w) = \eta[wg - (1-w)c]$. Escape from the lazy basin is governed not by drift but by its ratio to gradient noise,

$$\mathrm{SNR}(w) = \frac{D(w)}{\sqrt{\Theta(w)}} \gtrsim \kappa_c, \tag{2}$$

and the crossing defines $w_{\text{crit}}$. Because escape is a Kramers first-passage problem ($\tau \sim e^{\Delta U/\Theta}$), collapse switches abruptly across $w_{\text{crit}}$ (the sharp fits) and $w_{\text{crit}}(T)$ falls as $T$ grows (the metastable slide)—both predicted without fitting. We do *not* derive the numerical value of $w_{\text{crit}}$: $c, g, \sigma, \kappa_c$ are calibrated, and our preregistered direct drift probe was *inconclusive*—it reproduces the matched-horizon boundary in collapse fraction but does not localize $w_{\text{crit}}$, because survivor bias in the contested band confounds the conditioned drift (Supplement). The magnitude therefore remains calibrated; the mechanism's predictive content is the sign and the form.

The sign is a theorem.

**Theorem 1** (Sign invariance of the loss-aversion shift)**.** *Let the order parameter $x$ evolve as $\mathrm{d}x_t = F(x_t, w, \lambda)\,\mathrm{d}t + \sqrt{\Sigma(x_t, N)}\,\mathrm{d}W_t$ with $\Sigma > 0$. If $\partial_\lambda F < 0$ on the escape interval—i.e. loss aversion scales only the loss domain and so inflates gradient noise—then $\partial_\lambda w_{\text{crit}}^{\text{Emp}} > 0$.*

*Proof sketch.* The Kramers barrier $\Delta U = \int 2F/\Sigma\,\mathrm{d}z$ has $\partial_\lambda \Delta U < 0$ pointwise, since $\Sigma > 0$ and $\partial_\lambda F < 0$ throughout, independent of the sign of $F$; a lower barrier shortens escape ($\partial_\lambda \tau < 0$) while $\partial_w F > 0$ lengthens it. The implicit function theorem on $\tau(w, \lambda) = T$ gives

Table 1: Three-way decomposition of $w_{\text{crit}}$. We report the effect both ways: net vs. the linear null (combines curvature and loss aversion), and the pure loss-aversion increment vs. the curvature baseline (0.586). "Sig." = difference CI excludes zero.

| Condition | $w_{\text{crit}}$ | vs. null | vs. curv. |
|---|---|---|---|
| Null ($\lambda$=1) | 0.685 | — | — |
| Curvature ($\lambda$=1) | 0.586 | −0.100✓ | 0 |
| Loss av. $\lambda$=3 | 0.774 | +0.089✓ | +0.188 |
| Loss av. $\lambda$=7 | 0.851 | +0.166✓ | +0.265 |

$\mathrm{d}w_{\mathrm{crit}}^{\mathrm{Emp}}/\mathrm{d}\lambda = -\partial_\lambda \tau / \partial_w \tau > 0$. The magnitude depends on the intractable $\Sigma$; the direction does not. The same argument with curvature's drift signs reversed yields the opposite shift. □

We are candid that the theorem *encodes* the sign of $\partial_\lambda F$ as a premise rather than deriving it from first principles; its content is that this local sign propagates to the global boundary regardless of the intractable noise—which is exactly what makes the direction (but not the magnitude) predictable. The same drift split fixes the functional form: only the loss domain scales with $\lambda$, giving $w_{\mathrm{crit}}(\lambda) = \phi(\lambda)/(\phi(\lambda)+g/c)$ with $\phi(\lambda) = 1+\rho(\lambda-1)$ and $\rho \in [0,1]$ bounded by the theory. Fixing $g/c$ and $\rho = 0.61$ from a subset and testing out of sample, the one-parameter law tracks the full seven-point sweep to RMSE 0.014. This law is *single-substrate*: it does not transfer to LBF (§), a limit we report rather than hide.

## The Direction Generalizes; the Magnitudes Do Not

The mechanism commits in advance to a direction: loss aversion inflates gradient noise and so raises $w_{\mathrm{crit}}$. If that is right rather than an artifact of one substrate or optimizer, the sign should survive changing both. We test on two further environments and a second algorithm, matching the learner to each environment's learnability (independent PPO where coordination is locally learnable; centralized-critic MAPPO (Yu et al. 2022) where the reward is shared-global). The direction is robust in every setting; the absolute level, the curvature sign, and the functional form are not (Figure 3, Table 2). On the *supply chain under MAPPO* (only the learner changes), $w_{\mathrm{crit}}$ rises $0.796 \to 0.889$ over $\lambda{=}1 \to 7$, monotone from $\lambda{=}2$, endpoints non-overlapping; the level differs at low $\lambda$ but converges with PPO by $\lambda \geq 4$. On *MPE* the loss-aversion excess is $+0.158$ $[0.108, 0.212]$ at $\lambda{=}7$, but the curvature effect is null/opposite ($+0.042$, CI spanning zero), and on *LBF* curvature moves the boundary *right*—both confirming the curvature sign is substrate-specific. The supply chain's affine law misfits LBF by RMSE 0.15. Thus the *method*—a CI'd boundary that loss aversion shifts—generalizes across three environments and two algorithms, while the affine constants and the curvature sign do not.

Two ablations confirm the mechanism's scope conditions. Replacing the continuous policy gradient with value-based learners (QMIX, IQL) on a discretized action space erases the boundary—but so does discretizing PPO alone, isolating coarse discretization (not the value-based paradigm) as the cause, as the continuous-gradient mechanism predicts. And on shared-reward tasks the boundary is measurable only with a centralized critic: independent PPO collapses at *every* $w$ on MPE (a credit-assignment floor), while MAPPO recovers coordination. Both are detailed in the Supplement; each is a confirmed boundary condition, not a negative result.

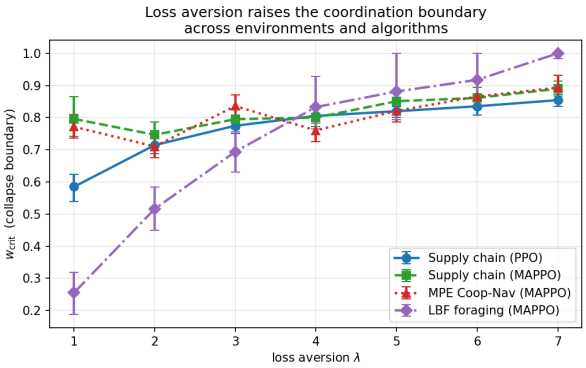



Figure 3: $w_{\mathrm{crit}}$ versus $\lambda$ across four settings: supply chain under PPO and MAPPO, MPE under MAPPO, LBF under MAPPO (bootstrap 95% CIs). $w_{\mathrm{crit}}$ rises with $\lambda$ in every setting; the *direction* generalizes, while levels (most at $\lambda{=}1$), curvature sign, and exact form are substrate-specific. LBF high-$\lambda$ points saturate against the ceiling and are exploratory.

Table 2: Generalization of the loss-aversion *direction*. The sign holds everywhere; only the supply chain (PPO) yields the affine law. Detail in the Supplement.

| Setting (algo.) | Loss-av. effect | Affine law? |
|---|---|---|
| Supply (PPO) | $+0.166$ ($\lambda{=}7$) | yes (0.014) |
| Supply (MAPPO) | $0.80 \to 0.89$ | — |
| MPE (MAPPO) | $+0.158$ ($\lambda{=}7$) | no |
| LBF (MAPPO) | $0.26 \to 0.83$ | no (0.15) |

## A Minority Sets the Boundary

The decomposition treats the population as homogeneous, but deployed systems mix loss-averse and risk-neutral participants. Assigning $\lambda{=}7$ to selected nodes and $\lambda{=}1$ to the rest (substrate, PPO config, label, grid identical to the main sweep), we find the boundary is set by a *minority*, and *network position* is decisive (Table 3). A single *downstream* loss-averse node drives $w_{\mathrm{crit}} = 0.853$ $[0.849, 0.858]$—statistically indistinguishable from a *fully* loss-averse population (0.851): one biased node downstream imposes the centralization requirement of the entire network, and two downstream nodes add nothing further. The same node *upstream* shifts the boundary far less, to 0.726 $[0.689, 0.763]$. Loss aversion thus cascades *upstream* along the ordering chain but not downstream, a directional asymmetry mirroring the substrate's bullwhip structure. That a behavioral minority can move a tipping point is established in evolutionary game theory (Mengesha, Sun, and Roy 2025); ours is the deep-MARL instantiation, where the effect emerges from gradient-learning dynamics and carries a concrete network-position signature.

Table 3: Heterogeneous-population sweep (loss-averse nodes at $\lambda$=7, others $\lambda$=1; $\alpha$=$\beta$=0.88, 300k, 20 seeds/cell). One downstream node already reaches the homogeneous loss-averse boundary; the cascade runs upstream, not down.

| Loss-averse nodes | $w_{\text{crit}}$ (95% CI) |
|---|---|
| one upstream | 0.726 [0.689, 0.763] |
| one downstream | 0.853 [0.849, 0.858] |
| two downstream | 0.851 [0.847, 0.856] |
| all five (homogeneous) | 0.851 [0.847, 0.856] |

The operational reading is sharp: in a mixed human–agent chain, the required centralization is set by the most downstream loss-averse participant, not the mean.

We are explicit about the limits of this result, which is the thinnest part of the paper. It uses binary $\lambda \in \{1, 7\}$ on one topology, and tests four configurations (one upstream, one downstream, two downstream, all); a continuous-$\lambda$ sweep and intermediate node positions are in progress to characterize the cascade as a function of position and bias strength.[2] The result is also single-substrate *by construction*: "which agent sets the boundary" is well-posed only where a directed resource flow defines upstream/downstream. MPE (spatial) and LBF (foraging) have none, so the question does not type-check there; a search for a second directed-flow cooperative-MARL substrate found none reachable (Supplement). We therefore establish the effect where its precondition holds and frame its reach as a *falsifiable, structurally specified* prediction: in any cooperative domain with a directed resource flow and a learning-reachable boundary, a downstream loss-averse minority should set it.

## Discussion

That probability weighting and loss aversion exert opposite effects on the boundary is, to our knowledge, novel; prior work applies prospect theory as a unified model without decomposing it (Lalmohammed et al. 2025; Ghaemi et al. 2024). The operational reading is concrete: a system of $\lambda$=7 agents needs $w > 0.851$ for reliable coordination against $w > 0.685$ for risk-neutral agents—a +0.166 deployment margin that scales with $\lambda$, so domains nearer $\lambda \approx$ 2–3 incur a smaller but positive cost. Where centralization is capped by privacy or communication cost, the boundary reframes the problem, exposing $\lambda$—the agents' effective loss aversion, shaped by the training objective—as a second lever. Our phase-diagram framing sits within a recognized line of multiagent phase-transition results (Yamaguchi 2025; Capannoli et al. 2026) and the broader project of cooperative AI (Dafoe et al. 2020; Hammond et al. 2025; Barfuss et al. 2020); its distinctive contribution is a boundary set by the policy-gradient signal-to-noise ratio rather than a fixed update rule.

**Limitations.** We state these plainly, as several bound the contribution. (i) The *quantitative* core—the affine $\lambda$-law, the metastable magnitudes, the cascade—is single-substrate; the affine law explicitly fails to transfer, and the cascade is one topology with binary $\lambda$. We replicate the boundary, the loss-aversion direction, and the metastable slide across environments and a second algorithm, but the precise numbers are a property of one simulator and one compute budget. (ii) The mechanism is calibrated, not predictive: it fixes sign and form, the direct drift probe was inconclusive, and the sign-invariance theorem encodes its key premise rather than deriving it. (iii) $\lambda$=7 is an upper, high-stakes bound rather than a typical human value ($\lambda \approx 2.25$); the deployment-relevant margins are at $\lambda \approx 3$. (iv) The high-$\lambda$ grid is coarse, and the horizon trend rests on two clean points; finer-grid and additional-horizon runs are in progress. None of these undercut the central, replicated claim—loss aversion raises the coordination requirement—but they scope what is measured versus what is predicted.

## Conclusion

We presented the first confidence-intervalled phase diagram of cooperative-MARL coordination collapse, characterizing $w_{\text{crit}}$ as a function of reward structure, training horizon, and prospect-theoretic loss aversion. Loss aversion impedes coordination—raising the requirement, a direction we predict a priori and confirm across four environment–algorithm settings, with the metastable slide generalizing likewise. Two further results sharpen the picture: prospect theory's two components act in opposite directions, and a downstream loss-averse minority can set the boundary on a directed-flow substrate, framed as a falsifiable prediction. Beyond the findings, the pre-registered phase-diagram protocol is reusable by any cooperative-MARL study. Ongoing work upgrades the mechanism from calibrated to predictive, tightens the high-$\lambda$ grid and horizon trend, and characterizes continuous heterogeneous-$\lambda$ populations whose tail the cascade predicts should govern the boundary.

[2]Runs varying node index along the chain and sweeping $\lambda$ continuously at a fixed downstream node are queued; this version reports the four binary configurations.

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